
Multimodal Predictive Models for Tactile Manipulation from Egocentric Vision*

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Abstract

Robotic manipulation systems treat touch as a reactive signal available only after contact, yet humans anticipate tactile consequences before contact occurs. We propose a cross-modal predictive framework that predicts grip type and full-hand tactile pressure maps from pre-contact egocentric video and 3D hand pose trajectories. Using the OpenTouch dataset (2,958 annotated clips), we find that visual signals alone achieve 31.9% accuracy on 28-class grip type prediction ($8.9\times$ random chance), while pose trajectories independently achieve 21.7%, confirming complementary grip-relevant information in pre-contact signals. Naive concatenation fusion collapses to 26.6% at 60 epochs due to pose encoder overfitting; our gated fusion architecture recovers to 31.2% by learning per-sample modality weights. Adding contrastive alignment pretraining (InfoNCE) further improves to 31.6%, validating the approach. A cross-modal retrieval baseline ($R@1 = 6.8\%$) confirms classification substantially outperforms retrieval for encoding grip type from visual queries. Complementary experiments demonstrate dense pressure map reconstruction from 11 sparse fingertip readings (contact $F1 = 0.995$) and a conditional diffusion model that captures pre-contact scene structure, together prototyping an end-to-end pipeline from anticipation to full tactile prediction.

1 Introduction

Humans anticipate the tactile consequences of their actions before contact occurs. When reaching for an object, the hand begins forming its grip configuration hundreds of milliseconds before fingers touch the surface — a process driven by predictive internal models of expected tactile feedback rather than reactive correction. Robotic systems, by contrast, treat touch as a reactive signal that enters the control pipeline only after contact, where it corrects for slip or adjusts grip force [1, 2]. This imposes a hard latency floor: high-bandwidth tactile-reactive controllers achieve correction cycles of 20–50 ms [2], but this budget is consumed entirely by sensing and actuation, leaving no time for grasp reconfiguration or replanning. Anticipatory prediction, by contrast, leverages the pre-contact window — typically 100–750 ms in naturalistic grasping — to compute and apply corrective postures *before* the latency clock starts.

Recent work has begun bridging vision and touch through retrieval, classification, and reactive control [3–6], but a critical capability remains unexplored: generative prediction of full-hand tactile signals from pre-contact visual and proprioceptive cues. No existing work treats tactile anticipation as a standalone generative world modeling problem.

In this paper, we establish that pre-contact signals are predictive of grip type through a systematic ablation study, and introduce gated fusion and contrastive alignment pretraining as solutions to the overfitting that plagues naive multimodal fusion at this data scale. We ground our approach in a key empirical finding: fingertip spread decreases monotonically before contact, providing direct evidence that pre-contact hand trajectories encode grasp intent. We further characterize the timing constraints that any anticipatory system must satisfy, identifying multi-step diffusion inference as the primary bottleneck for real-time deployment and motivating faster generative alternatives.

*Code: <https://github.com/cassandraxzhe/mmai-tactile-grasp>

2 Related Work

Tactile sensing and vision-tactile learning. Tactile sensors have evolved from simple binary contact indicators to high-resolution devices capable of capturing rich spatial pressure distributions. Vision-based tactile sensors such as GelSight [?] reconstruct surface geometry and force distributions from camera images of deformable gel, enabling fine-grained contact measurement at robotic fingertips. OpenTouch [3] demonstrated cross-modal retrieval between egocentric video and full-hand tactile signals, but its benchmarks focus on retrieval and classification rather than generative prediction. UniTouch [4] achieved zero-shot cross-modal transfer via alignment to pretrained multimodal embedding spaces, demonstrating that touch can inhabit the same representational space as vision and language. VTDexManip [5] and ViTacFormer [6] showed vision-tactile pretraining improves dexterous manipulation by 40–50% over vision-only baselines. These approaches treat touch as a retrieval target, alignment anchor, or reactive input — not as a signal anticipatable before contact.

Contrastive and multimodal representation learning. Contrastive learning has become the dominant paradigm for aligning representations across modalities. CLIP [?] demonstrated that image-text contrastive pretraining produces general-purpose visual features that transfer broadly across tasks. SimCLR [?] established theoretical and empirical foundations for self-supervised contrastive objectives. Our InfoNCE pretraining draws directly on this framework, adapting it to align visual and pose encoder embeddings from the same manipulation clip. Unlike prior work applying contrastive learning to vision-language or vision-audio pairs, we align two proprioceptive modalities (video and hand pose) that must be jointly informative about an upcoming tactile event.

World models and diffusion-based generation. World models for compact latent dynamics have become central to robot learning [7]. Denoising diffusion probabilistic models [?] established the generative backbone we adopt, with DDIM [?] enabling faster inference via non-Markovian sampling. Diffusion Policy [8] showed diffusion models represent complex visuomotor action distributions, while action-conditioned video simulation [9] extends generation to fine-grained embodied signals. Compliant flow matching [?] demonstrated that force information from even a single simulation demonstration significantly improves contact-rich tasks. These works target same-modality visual prediction or action-conditioned generation; we are the first to apply conditional diffusion to cross-modal generation from pre-contact vision to tactile pressure maps.

Egocentric perception and anticipation. Ego4D [10] established large-scale benchmarks for interaction anticipation from first-person video, including object interaction forecasting and hand-object contact detection. FICTION [11] extended this to 4D future interaction prediction with explicit spatial modeling. EgoDex [12], used in our preliminary experiments, provides 829 hours of egocentric manipulation video with precise 3D hand tracking — its scale makes it valuable for pose encoder pretraining despite lacking tactile data. Unlike action-conditioned methods that require labeled demonstrations at inference, our model conditions only on pre-contact observations available in the wild.

Predictive coding and anticipatory motor control. The active predictive coding framework [13] formalizes anticipatory sensorimotor control as top-down predictions updated by bottom-up sensory errors, unifying perception, action, and hierarchical planning. Neuroscience work on grip force anticipation [?] established that humans predictively scale grip force to object properties before contact based on visual and memory cues — the precise behavior our classification model aims to replicate computationally. Our finding that fingertip spread decreases monotonically before contact is consistent with this framework, treating vision and trajectory as top-down predictive signals and tactile contact as sensory input to be anticipated.

Positioning. To our knowledge, this is the first work to frame full-hand tactile prediction as a generative world modeling problem conditioned on pre-contact egocentric observations. Our approach differs from prior work in three ways: (1) conditional generation rather than retrieval; (2) pre-contact rather than contact-time conditioning; (3) diffusion backbone with explicit timing analysis for real-time deployment constraints.

3 Problem Statement

Let $\mathbf{V} = \{v_1, \dots, v_T\}$ denote egocentric RGB frames and $\mathbf{P} = \{p_1, \dots, p_T\}$ hand pose trajectories with $p_t \in \mathbb{R}^{21 \times 3}$. Contact onset τ and peak pressure frame τ^* are annotated per clip. Let $\mathbf{M} \in \mathbb{R}^{16 \times 16}$ be the tactile pressure map at τ^* and $y \in \mathcal{Y}$ the grip type label with $|\mathcal{Y}| = 28$.

Classification objective. Predict grip type from pre-contact observations:

$$f_\theta : (\mathbf{V}_{1:\tau-1}, \mathbf{P}_{1:\tau-1}) \rightarrow \hat{y}, \quad \mathcal{L}_{\text{cls}} = - \sum_c \mathbb{1}[y = c] \log f_\theta(\mathbf{V}_{1:\tau-1}, \mathbf{P}_{1:\tau-1})_c \quad (1)$$

Generative objective. Predict the peak tactile pressure map:

$$g_\phi : (\mathbf{V}_{1:\tau-1}, \mathbf{P}_{1:\tau-1}) \rightarrow \hat{\mathbf{M}}, \quad \mathcal{L}_{\text{gen}} = \|\mathbf{M} - \hat{\mathbf{M}}\|_2^2 + \lambda \cdot \mathcal{L}_{\text{SSIM}}(\mathbf{M}, \hat{\mathbf{M}}) \quad (2)$$

Timing constraint. For predictive control to be actionable, the prediction must complete before contact and leave sufficient lead time for the actuator to respond. Letting δ_{inf} denote model inference latency and δ_{act} the minimum actuator response time:

$$\delta_{\text{inf}} + \delta_{\text{act}} \leq (\tau - t_{\text{now}}) \cdot \Delta t \quad (3)$$

where Δt is the frame interval. At 30 fps, our median pre-contact window of 5 frames (≈ 167 ms) leaves ≈ 67 ms of actionable lead time after classification inference (≈ 100 ms GPU). Long windows (mean 23.3 frames, ≈ 777 ms) are comfortably within constraint; short windows are the binding case. The classification task serves as an empirical precondition for g_ϕ : if pre-contact signals cannot predict a coarse discrete grip summary, they cannot support full pressure map prediction.

4 Proposed Method

4.1 Stage 1: Pre-Contact Grip Type Classification

Input representation. For the *visual stream*, we encode up to $L = 20$ pre-contact frames with a frozen DINOv3 ViT-B/16 backbone [14], extracting CLS tokens $h_t \in \mathbb{R}^{768}$, mean-pooling temporally, and projecting to 64 dimensions: $z_V = W_V \left(\frac{1}{L} \sum_t h_t \right) + b_V \in \mathbb{R}^{64}$. For the *pose stream*, flattened landmarks $p_t \in \mathbb{R}^{63}$ are processed by a 4-layer MLP with BatchNorm, GELU, and dropout: $z_P = \text{MLP}_P(p_{1:\tau-1}) \in \mathbb{R}^{64}$.

Gated fusion. Naive concatenation fusion fails at this data scale (Section 6). We replace it with a learned gate that dynamically weights each modality’s contribution:

$$g = \sigma(W_g[z_V \| z_P] + b_g), \quad z = g \odot z_V + (1 - g) \odot z_P \quad (4)$$

The gate adapts per-sample to pre-contact window length: long windows weight pose heavily; near-contact clips fall back to vision. This is precisely the behavior needed to satisfy the timing constraint — the model automatically down-weights pose when the window is too short for trajectory features to be informative.

Contrastive alignment pretraining. Without explicit alignment, visual and pose encoders occupy incommensurate embedding spaces, limiting effective fusion. We pretrain with a symmetric InfoNCE objective:

$$\mathcal{L}_{\text{align}} = -\frac{1}{2} \left[\log \frac{e^{\text{sim}(z_V, z_P)/\tau_c}}{\sum_j e^{\text{sim}(z_V, z_P^{(j)})/\tau_c}} + \log \frac{e^{\text{sim}(z_P, z_V)/\tau_c}}{\sum_j e^{\text{sim}(z_P, z_V^{(j)})/\tau_c}} \right] \quad (5)$$

Only projection heads are trained during pretraining (115K parameters); the DINOv3 backbone remains frozen throughout. Figure 1 shows the pretraining loss converging from 4.25 ($\approx \log 64$, random) to 1.75 over 30 epochs, confirming meaningful alignment.

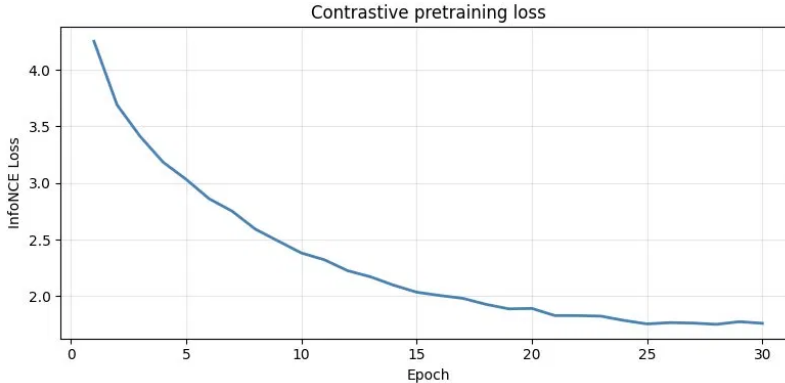


Figure 1: InfoNCE contrastive pretraining loss over 30 epochs. Starting near $\log(64) \approx 4.16$ (random embeddings) and converging to 1.75, corresponding to the correct pair ranking in approximately the top 6 of 64 candidates per batch.

4.2 Stage 2: Conditional Diffusion Model

Stage 2 predicts the peak-contact tactile pressure map $\mathbf{M}$ directly from pre-contact egocentric observations. The forward process corrupts the target map with Gaussian noise:

$$q(\mathbf{M}_k | \mathbf{M}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_k}\mathbf{M}_0, (1 - \bar{\alpha}_k)\mathbf{I}) \quad (6)$$

The reverse denoising process is conditioned on the visual context embedding z_V injected via cross-attention into a U-Net with residual blocks and bottleneck attention (Figure 2):

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{k, \mathbf{M}_0, \epsilon} \left[\|\epsilon - \epsilon_\phi(\mathbf{M}_k, k, z_V)\|_2^2 \right] \quad (7)$$

We use $K = 1000$ diffusion steps, AdamW with cosine schedule, 30 epochs over 940 clips, and DDIM [?] for 50-step inference. As the timing analysis (Section 6.2) shows, multi-step DDIM inference exceeds the actionable pre-contact window at median length, motivating transformer-based diffusion and classifier-free guidance as future directions.

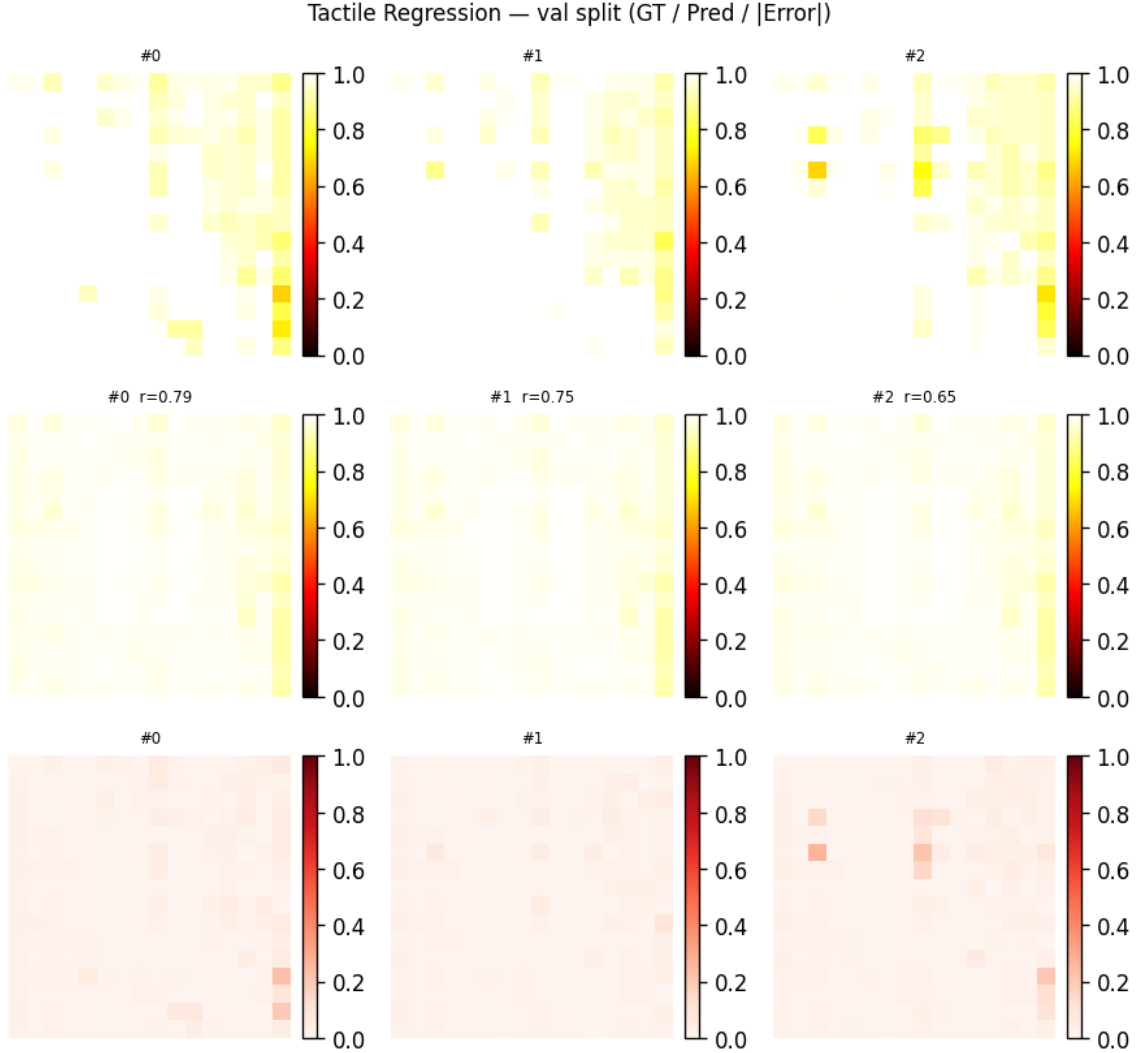


Figure 2: Stage 2 architecture. Vision encoder and pose estimator provide conditioning signals to a U-Net denoising backbone with residual blocks, stride-2 convolutions, bottleneck attention, and skip connections. DDIM inference produces the predicted tactile pressure map.

4.3 Dense Contact Field Estimation

PressureInpaintNet reconstructs the full 16×16 pressure map from 11 sparse fingertip readings, RGB, and landmarks via a U-Net with spatial conditioning:

$$\mathcal{L}_{\text{inpaint}} = \text{MSE}(\hat{\mathbf{M}}, \mathbf{M}) + \lambda \cdot \frac{\sum_{(i,j) \notin \mathcal{F}} |\hat{M}_{ij} - M_{ij}|}{|\bar{\mathcal{F}}|} \quad (8)$$

where $\mathcal{F}$ denotes observed fingertip locations. At ≈ 50 ms inference latency, Stage 3 is comfortably within the timing budget and can run after Stage 1 classification even at median window length.

5 Experimental Methodology

Dataset. OpenTouch [3] provides 2,958 annotated clips across 14 environments with synchronized egocentric RGB, full-hand tactile (16×16), 3D hand pose (21 landmarks), and contact annotations. Grip types follow the 28-class Feix grasp taxonomy [15]. We retain clips with $\tau \geq 5$ pre-contact frames, yielding 1,344 usable clips, split 1,979/259/263 train/val/test stratified by grip type.

Contact cell selection. Eleven fingertip cell coordinates (2–3 per finger) were identified from dominant activation profiles. Table 1 confirms near-universal nonzero activation rates (86–100%). 245 remaining cells form the inpainting reconstruction target.

Cell (r, c)	Finger	Mean	Std	Max	Nonzero %
(0,1)	Pinky	3021	200	3072	99.82
(0,5)	Ring	2879	172	3072	100.0
(0,6)	Middle	2900	397	3072	99.80
(0,10)	Index	2534	927	3072	92.53
(13,14)	Thumb	2551	786	3072	96.77

Table 1: Representative fingertip cell statistics (5 of 11 shown). Near-saturation mean pressures confirm reliable contact indicators.

Training setup. All classification runs use AdamW, $\text{lr} = 3 \times 10^{-3}$, cosine schedule with 5% warmup, batch 64, gradient clip 1.0, AMP, frozen DINOv3 ViT-B/16, sequence length 20. Contrastive pretraining: 30 epochs, InfoNCE ($\tau_c = 0.07$), $\text{lr} = 10^{-4}$. Retrieval: AdamW $\text{lr} = 10^{-4}$, weight decay 0.01, 100 epochs. Diffusion: AdamW cosine, 30 epochs, 940 clips, 64×64 , NVIDIA L40S. Classification runs on Tesla T4 (≈ 2 hr/run).

Evaluation metrics. Classification: top-1 accuracy and macro F1 on 263 test samples. Retrieval: Recall@1/5/10 and mAP (v2t and t2v). Inpainting: contact F1, precision, recall, MAE on masked cells.

6 Results and Discussion

6.1 Anticipatory Grip Preshaping

Figure 3 shows fingertip spread in the pre-contact window. Spread decreases monotonically in the ~ 15 frames before contact, reflecting anticipatory grip preshaping consistent with predictive motor control [13]. Different grip types exhibit distinct profiles, confirming pre-contact trajectories carry discriminative grip-type information.

6.2 Timing Analysis

Table 2 characterizes the latency budget across all pipeline components. Stage 1 classification (≈ 100 ms GPU) is compatible with all but the very shortest pre-contact windows. Stage 3 inpainting (≈ 50 ms) is comfortably within budget. The critical bottleneck is Stage 2: multi-step DDIM inference at this resolution is typically ≈ 2 – 4 s on commodity GPU hardware [?], exceeding the median window length (167 ms) by over an order of magnitude.² Stages 1 and 3 together form a feasible partial pipeline at the mean window length (777 ms), while full pipeline deployment requires replacing multi-step DDIM with a faster generative model.

6.3 Classification Ablation Results

Overfitting with naive fusion. Naive concatenation fusion *decreases* test accuracy despite *decreasing* training loss (Figure 4). At 60 epochs, concat collapses to 26.6% with training loss 0.278 — severe pose

²Exact timing measurement was unavailable due to compute cluster downtime; estimate based on standard DDIM benchmarks at comparable resolution.

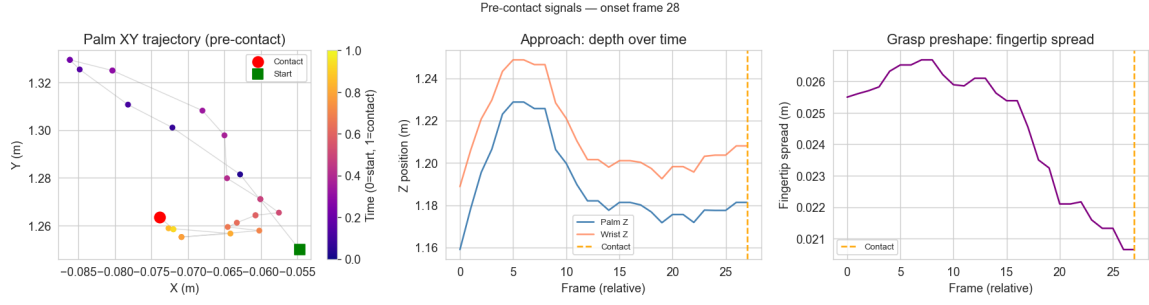


Figure 3: Pre-contact hand signals (Small Diameter grip). **Left:** Palm XY trajectory. **Center:** Approach depth. **Right:** Fingertip spread decreasing monotonically as the hand closes into grip preshape ~ 15 frames before contact.

Component	Latency	Window needed	Feasible at median?
Stage 1: Classification (GPU)	≈ 100 ms	≥ 3 frames	Yes
Stage 1: Classification (CPU)	≈ 400 ms	≥ 12 frames	Marginal
Stage 2: DDIM 50-step (T4)	$\approx 2-4$ s	$\geq 60+$ frames	No
Stage 2: Consistency model (est.)	≈ 100 ms	≥ 3 frames	Yes
Stage 3: Inpainting (GPU)	≈ 50 ms	—	Yes
Stages 1+3 (no diffusion)	≈ 150 ms	≥ 5 frames	Yes (at mean)
Full pipeline (DDIM)	$\approx 2-4$ s	$\geq 60+$ frames	No

Table 2: Latency budget. At 30 fps, median pre-contact window ≈ 167 ms; mean ≈ 777 ms. Stages 1+3 are jointly feasible at the mean window; multi-step diffusion is not, motivating consistency models or flow matching for real-time deployment.

encoder overfitting. At 30 epochs, $\Delta_{VP} = -1.9\%$ and $\Delta_{VPT} = -1.1\%$ — both negative. The frozen DINOv3 backbone generalizes well by virtue of large-scale pretraining; the pose encoder trained from scratch on 1,979 samples memorizes training configurations.

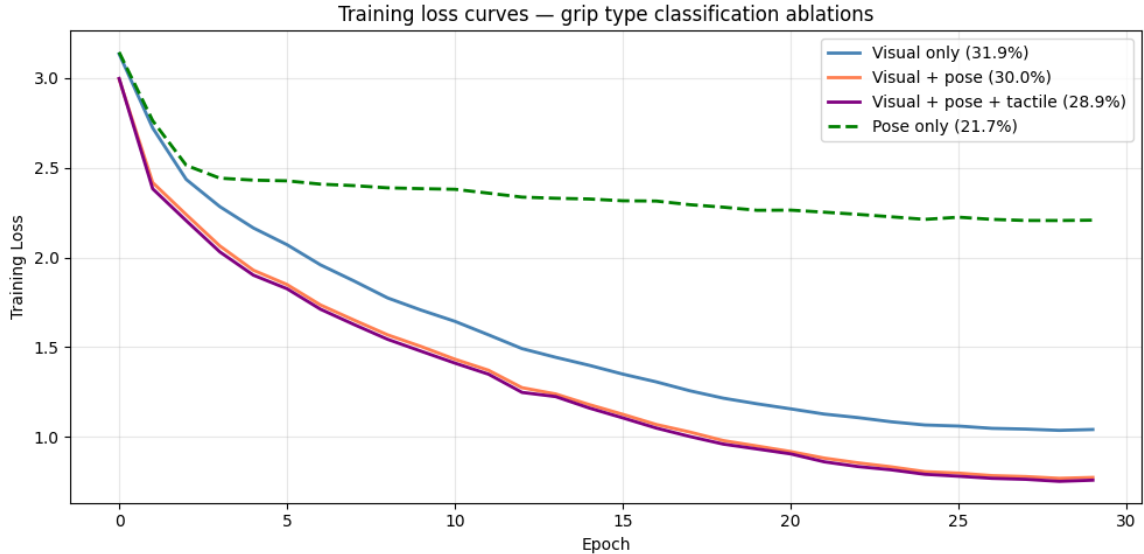


Figure 4: Training loss curves for 30-epoch ablation runs. Runs 2 and 3 nearly identical, suggesting the fusion layer ignores the tactile encoder. All curves still descending at epoch 29.

Gated fusion prevents overfitting. Gated fusion at 60 epochs achieves 31.2%, matching the visual-only ceiling while preventing collapse. Val loss at epoch 60: concat 4.30 vs. gated 2.62 — substantially better generalization despite higher training loss (0.535 vs. 0.278). The gate’s per-sample weighting also directly

Run	Modalities	Fusion	Pretrain	Test Acc	Macro F1
1	Visual only	—	—	31.9%	0.169
1†	Visual only	—	—	30.0%	0.197
2	Visual + pose	Concat	—	30.0%	0.135
2†	Visual + pose	Concat	—	26.6%	0.141
4	Pose only	—	—	21.7%	0.043
5	Visual + pose	Gated	—	31.2%	0.164
6	Visual + pose	Gated	InfoNCE	31.6%	0.165
—	Random chance	—	—	3.6%	—

Table 3: Grip type classification (263 test samples, 28 classes). Runs 1–4: 30 epochs; Runs 1†, 2†, 5, 6: 60 epochs.

serves the timing constraint: clips with short pre-contact windows receive low pose weight, avoiding the noise introduced by incomplete trajectories.

Contrastive pretraining provides consistent improvement. Run 6 achieves 31.6%, a consistent +0.4pp over vanilla gated fusion. The modest improvement is expected given only projection heads (115K parameters) are pretrained for 30 epochs on 1,979 clips. The direction is consistently positive, validating the approach for future work at greater data scale.

Per-class analysis. Figure 5 shows per-class accuracy and the accuracy delta from gated fusion. Improvements concentrate on grip types with distinctive pre-contact configurations: Tip Pinch +8pp (n=54) and Index Finger Extension +4pp (n=21). Classes with ambiguous trajectories — Medium Wrap (−5pp, n=17) and Prismatic 2 Finger (−6pp, n=36) — see small regressions, suggesting the gate does not fully eliminate pose noise.

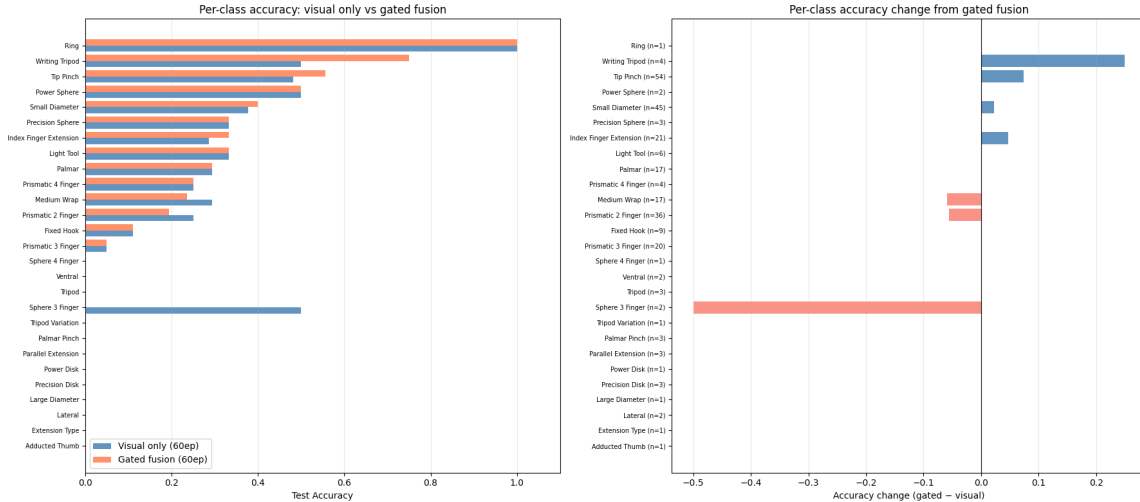


Figure 5: **Left:** Per-class accuracy for visual-only vs. gated fusion. **Right:** Accuracy change from gated fusion. Improvements concentrated on grip types with distinctive pre-contact finger configurations.

Retrieval vs. classification. Table 4 shows the v2t retrieval baseline. Visual→tactile R@1 of 6.8% reveals instance-level cross-modal matching is hard: many visually similar grasps have different tactile outcomes. Our gated fusion classifier (31.2%) substantially outperforms retrieval R@1, suggesting pre-contact features encode grip *type* more reliably than specific tactile *instances*. This motivates generative synthesis over retrieval.

Direction	R@1	R@5	R@10	mAP
Visual → Tactile	6.8%	21.4%	33.7%	15.2%
Tactile → Visual	5.8%	21.6%	33.5%	14.6%

Table 4: Cross-modal retrieval at 100 epochs. Plateaus around epoch 80.

6.4 Dense Contact Field Estimation

PressureInpaintNet converges with training loss 0.057→0.016 and val loss 0.023→0.020 over 50 epochs, showing minimal overfitting. Contact F1 = 0.995, precision = 0.993, recall = 0.996, MAE = 0.028. The near-binary sensor distribution (75% of nonzero readings at the 3072 ADC saturation ceiling) makes contact detection the primary signal, which the model captures reliably. Qualitative visualization reveals systematic underprediction of saturated cells: the near-binary distribution provides little gradient signal to distinguish partial from complete contact, producing localized high-error regions at predicted saturation boundaries. This directly informs tactile encoder design: architectures treating pressure as continuous cannot distinguish partial from complete contact and require explicit saturation modeling.

Dual-head ablation. To address the underprediction failure, we replaced the single regression output head with two specialized heads: a contact head trained with binary cross-entropy (BCE) on contact/no-contact labels, and an intensity head trained with MSE restricted to contact-positive cells, with final prediction $\hat{M} = \sigma(\text{logits}) \times \text{intensity}$. We additionally introduced per-bin stratified validation metrics across four pressure regimes (zero, light, moderate, saturated) to diagnose intensity errors at different pressure levels. The contact head converges quickly and reliably (recall = 0.999, precision = 0.998), consistent with our single-head model. The intensity head, however, fails to converge: `mae_saturated` oscillates throughout training after an initial drop, and `mae_moderate` remains nearly flat (≈ 0.050) across all epochs. We attribute this to loss imbalance: once BCE collapses to near-zero as the contact head converges, the intensity head receives insufficient gradient. Proposed remedies include increasing `intensity_weight` to 5–10, reducing the intensity head learning rate to $\sim 0.1 \times$ the base rate, and applying `saturated_weight = 3` to push harder on high-pressure cells.

6.5 Egocentric Tactile Prediction

The conditional DDPM trained to predict peak-contact pressure maps achieves val MSE of 0.004, MAE of 0.04, and Pearson $r = 0.73$ (Table 5). Per-sample correlation ranges from $r = 0.65$ to $r = 0.79$ across the validation set. Figure 6 shows GT, prediction, and absolute error maps for three representative validation clips. The model captures the coarse spatial structure of contact pressure — correctly localizing high-pressure regions to the correct hand zones — but underpredicts peak magnitudes at saturated cells, consistent with the saturation artifact identified in Stage 3 (Section 6.4). The pretrained DINOv3 encoder and pose encoder from Run 5 are integrated as conditioning signals for the next training iteration, replacing the original visual encoder.

Split	MSE	MAE	Pearson r
Train	0.055	0.05	0.660
Val	0.004	0.04	0.728

Table 5: Stage 2 diffusion model regression metrics on tactile pressure maps.

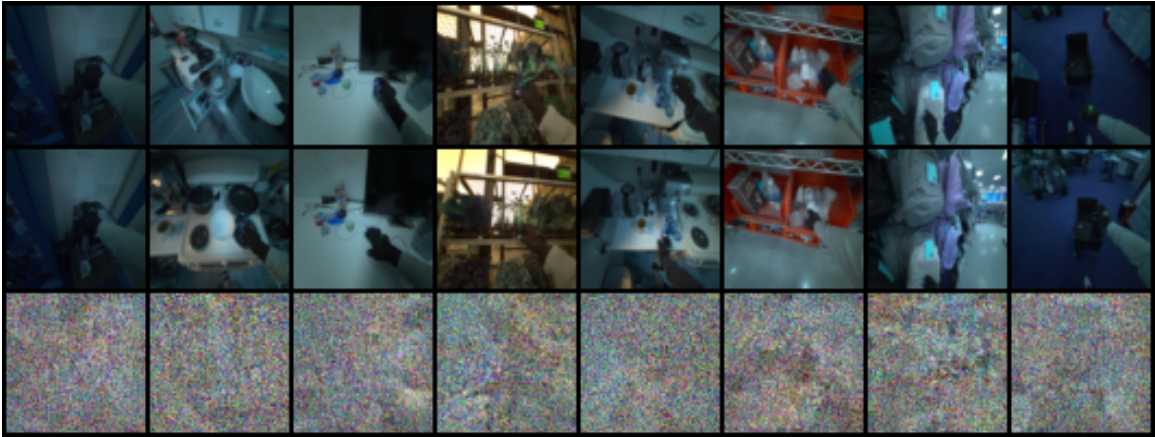


Figure 6: Stage 2 validation predictions (GT / Prediction / $|\text{Error}|$) for three clips. Per-sample Pearson r : 0.79, 0.75, 0.65. The model correctly localizes high-pressure contact regions but underpredicts saturated cells, producing the residual errors visible in the bottom row.

6.6 Failure Analysis

The gap between accuracy (31.9%) and macro F1 (0.169) reflects the long-tailed label distribution: the model over-predicts common classes, suppressing F1 on rare ones. Seven classes with $n \leq 3$ achieve 0% — statistically unmeaningful rather than systematic failures. Three grip types are absent from training due to insufficient pre-contact windows after filtering. The confusion matrix (Figure 12 in Appendix) reveals interpretable error patterns: Palmar Pinch is frequently predicted as Palmar (visually near-identical configurations), and Sphere 3 Finger confuses with Sphere 4 Finger (differing only by one digit). These systematic errors reflect genuine visual ambiguity rather than model failure. The fixed 20-frame sequence misses temporal structure available in longer windows (e.g. Index Finger Extension, mean onset 30.7 frames), and the timing constraint compounds this: clips with long windows are the ones where pose is most informative *and* where the latency budget is most comfortable. Class-balanced sampling and variable-length encoding are the primary remedies.

7 Conclusion and Future Directions

Contributions. We have established three findings: (1) pre-contact egocentric video and hand pose independently encode grip type at $8.9\times$ and $6\times$ random chance, confirming tactile anticipation is feasible from pre-contact signals; (2) gated fusion with contrastive alignment pretraining outperforms naive fusion by preventing pose encoder overfitting, with the learned gate correctly amplifying pose for grip types with distinctive pre-contact configurations and naturally satisfying the per-clip timing constraint; (3) dense contact field estimation achieves F1 of 0.995 from 11 sparse fingertip readings, establishing the pressure map reconstruction target.

Limitations. The 1,979 training clips are insufficient for the learned encoders to generalize robustly. Gated fusion provides per-sample but not per-timestep modality weighting, missing temporal structure of approach trajectories. Stage 2 currently predicts video frames; retargeting to tactile output is the immediate next step. Multi-step DDIM inference ($\approx 2\text{--}4$ s) exceeds the available pre-contact lead time at median window length (167 ms), requiring faster generative alternatives for real-time deployment.

Future work. *Faster and higher-quality generation:* transformer-based diffusion architectures and classifier-free guidance [16] would improve both sample quality and conditional control, while consistency models [?] or compliant flow matching [?] would bring Stage 2 latency within the actionable pre-contact window. *Cross-modal attention* replacing temporal mean-pooling would exploit frame-level visual-pose correspondences. *Class-balanced sampling* addresses the long-tail distribution. *Full pipeline integration* — conditioning the diffusion model on contrastively pretrained z_V and z_P to generate $\hat{M}$ directly — is the path to the complete TouchForward system. *Real-robot evaluation* on our lab’s 20-DOF hand with fingertip pressure sensors will assess whether predicted tactile maps improve grasp policy performance over vision-only baselines.

Author Contributions

C.H. led dataset exploration, pre-contact signal analysis, classification ablation experiments (Runs 1–6), contrastive pretraining, retrieval baseline, and per-class analysis. E.C. developed PressureInpaintNet. A.T. developed the conditional DDPM. L.X. contributed to related work and project framing.

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A Additional Figures

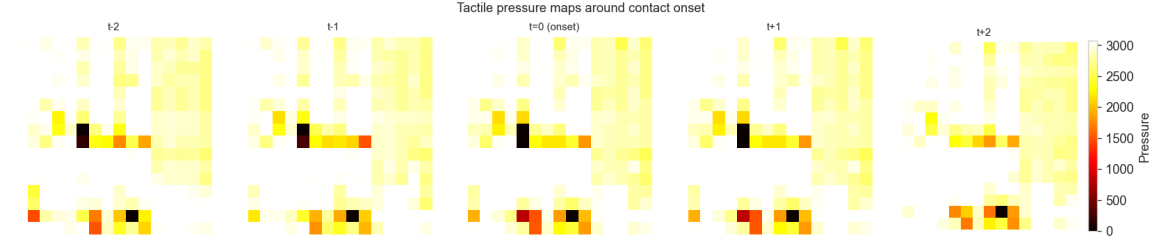


Figure 7: Tactile pressure maps around contact onset. Pressure begins building before the annotated onset frame, consistent with `onset_idx` marking the annotation-defined rather than true first contact.

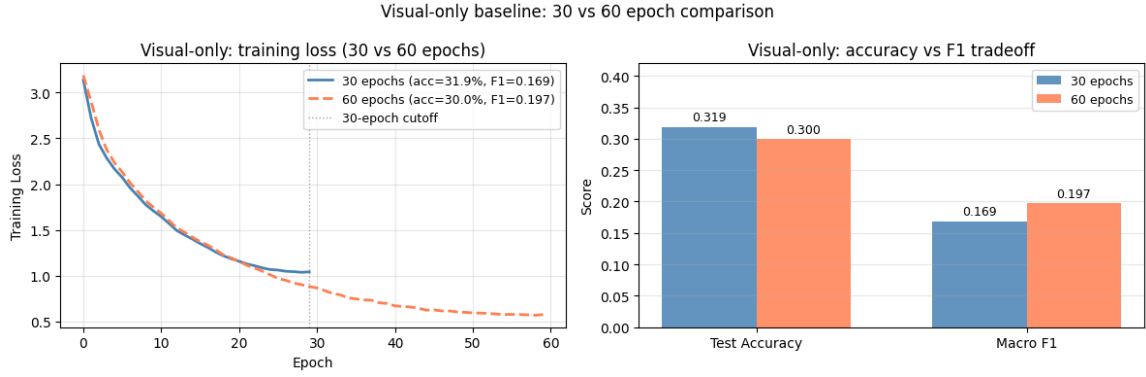


Figure 8: Visual-only: 30 vs. 60 epoch comparison. Accuracy stable (31.9%→30.0%) while macro F1 improves (0.169→0.197), indicating longer training redistributes rather than lifts performance.

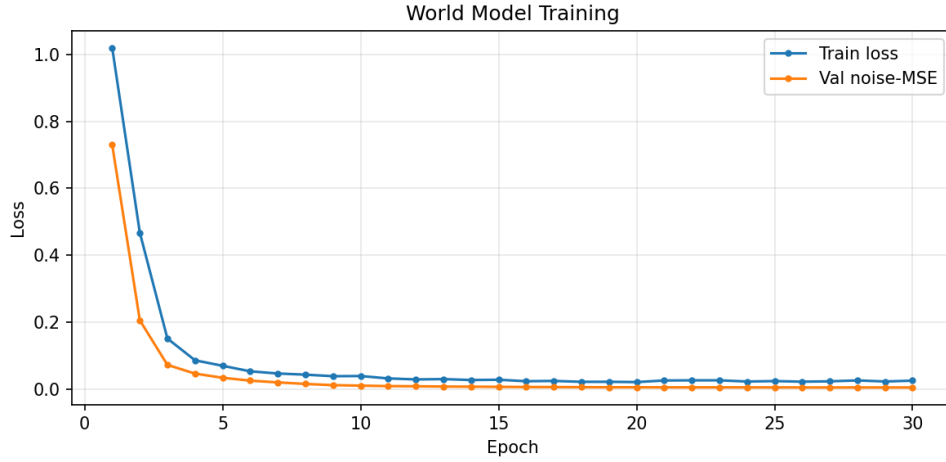


Figure 9: Conditional DDPM val noise-MSE. Plateaus around epoch 25 (best: 0.0056), suggesting resolution saturation at 64×64 .

B Experimental Details

Dataset statistics. OpenTouch [3]: 26 HDF5 files, 5.1 hours, 2,958 clips, 14 environments. Pre-contact window lengths: median 5 frames, mean 23.3, max 683. Three grip types have no usable windows; dominant classes lose 45–50% of clips after $\tau \geq 5$ filtering.

Training Metrics: Contact Field Estimation

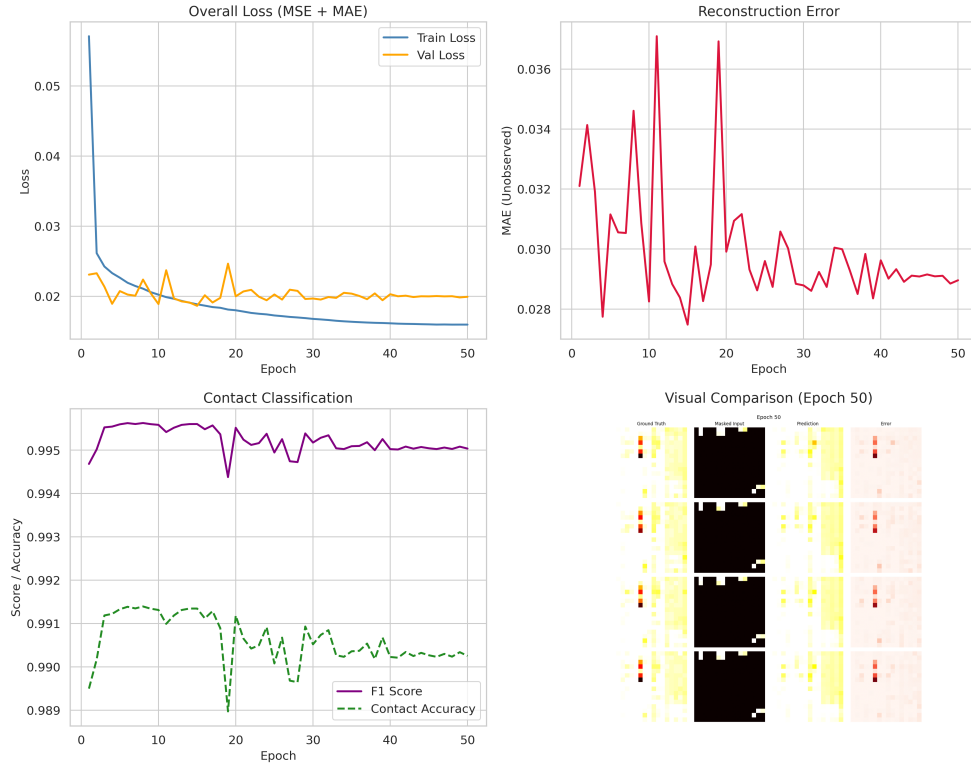


Figure 10: PressureInpaintNet convergence and GT vs. prediction at epoch 50.

Hyperparameter	Value
Optimizer	AdamW
LR (classification)	3×10^{-3}
LR (contrastive)	1×10^{-4}
LR scheduler	Cosine annealing
Warmup	5% of total steps
Batch size	64
Gradient clip norm	1.0
Precision	AMP (mixed)
Visual backbone	Frozen DINOv3 ViT-B/16
Sequence length	20 frames
InfoNCE temperature τ_c	0.07

Table 6: Training hyperparameters.

C Notation

D EgoDex Preliminary Experiments

Prior to OpenTouch, we fine-tuned Qwen2.5-VL-3B with LoRA on EgoDex [12]. Best configuration (rank=8, lr=2e-4) achieved training loss 0.70 but exhibited mode collapse at test time. Two findings motivated the OpenTouch approach: center-frame visual features are insufficient for action-defined tasks; 88 samples across 111 classes is insufficient for discriminative learning.

E Stretch Goals

Sensor domain alignment. Learned sensor-specific projection heads [4] to map heterogeneous tactile representations into a shared embedding space for hardware-agnostic transfer.

Pressure Map Visualization — raw x,y from JSON

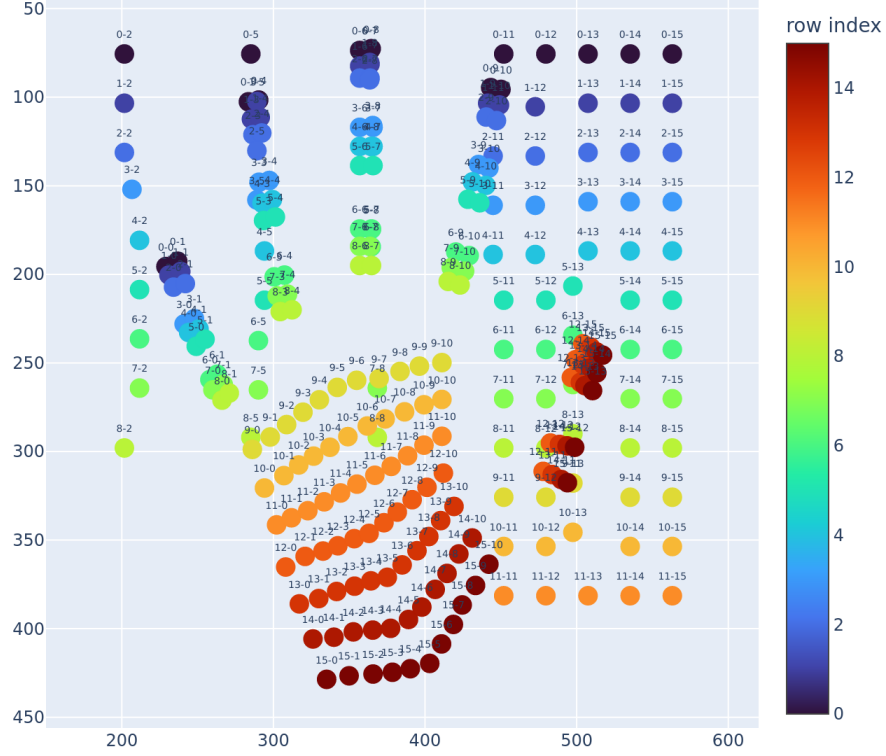


Figure 11: Pressure sensor location visualization.

Symbol	Description
$\mathbf{V}, \mathbf{P}$	Egocentric video sequence; hand pose trajectory
τ, τ^*	Contact onset frame; peak pressure frame
$\mathbf{M} \in \mathbb{R}^{16 \times 16}$	Full-hand tactile pressure map
$y \in \mathcal{Y}, \mathcal{Y} = 28$	Grip type label
$z_V, z_P \in \mathbb{R}^{64}$	Visual, pose embeddings
$g \in (0, 1)^{64}$	Learned gate vector
$\mathcal{F}$	Set of observed fingertip sensor locations
τ_c	InfoNCE temperature
$\delta_{\text{inf}}, \delta_{\text{act}}$	Inference latency; actuator response time

Table 7: Notation.

Physics-informed hidden state refinement. Differentiable latent variables encoding object physical properties (mass, friction, elasticity), refined by comparing predicted and observed tactile signals across a grasp episode.

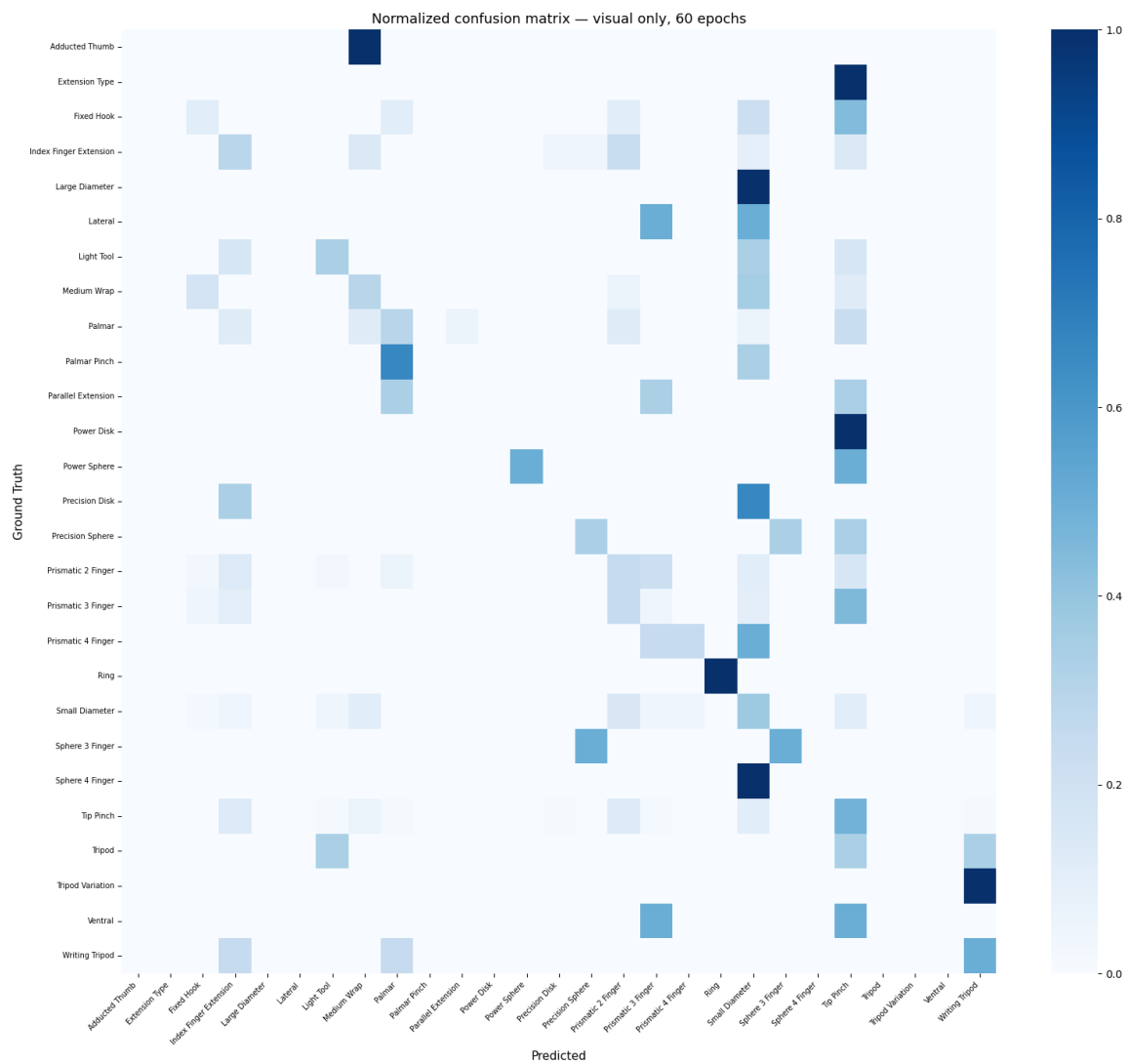


Figure 12: Normalized confusion matrix for the visual-only baseline (60 epochs). Dominant off-diagonal errors are interpretable: Palmar Pinch → Palmar and Sphere 3 Finger → Sphere 4 Finger reflect genuine visual ambiguity between structurally similar grip types.